Fact Sheet

US MILITARY AND INTELLIGENCE RENDEZVOUS AND PROXIMITY OPERATIONS

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SUMMARY

Over the last forty years, the United States has developed significant capabilities for conducting robotic rendezvous and proximity operations (RPO) to support military and intelligence-related space activities. The US military and intelligence services have flown multiple satellites to develop, practice, and utilize RPO technologies for close surveillance and inspection of other space objects as well as collecting signals intelligence. Some of these missions have been publicly acknowledged but most remain shrouded in secrecy, with a few completely unacknowledged publicly. None of the programs listed here have strong evidence to link them to co-orbital anti-satellite (ASAT) testing or deployment.

RENDEZVOUS & PROXIMITY OPERATIONS

Proximity operations are a series of orbital maneuvers executed to place and maintain a spacecraft in the vicinity of another space object on a relative planned path for a specific time duration to accomplish mission objectives. Rendezvous is a process whereby two space objects, artificial or natural bodies, are intentionally brought close together through a series of orbital maneuvers at a planned time and place. Taken together, RPO technologies enable a wide range of capabilities to support civil and commercial space activities such as on-orbit inspections, repair, refueling, assembly, and life extension. RPO capabilities can also be used for military and intelligence space activities such as intelligence, surveillance, and offensive weapons such as co-orbital anti-satellites. Since the end of the Cold War, the US Air Force (USAF), National Aeronautics and Space Administration (NASA), and Defense Advanced Research Projects Agency (DARPA) have all conducted tests and demonstrations of robotic close approach and rendezvous technologies in low and geosynchronous Earth orbits (LEO and GEO).

US RPO FOR MILITARY AND INTELLIGENCE MISSIONS IN LEO

On January 29, 2003, the USAF launched the XSS-10 as a secondary payload on a Delta-2 rocket carrying a US military GPS satellite. XSS-10 conducted a pre-planned series of RPO maneuvers near the Delta upper stage, eventually closing to within 50 m.[1] Similarly, XSS-11 was launched on April 11, 2005, and over the following 12 to 18 months, the spacecraft "conduct[ed] [RPO] maneuvers with several US-owned, dead or inactive resident space objects near its orbit."[2] However, it is impossible to verify whether or not these activities occurred, and whether or not XSS-11 visited any non-US space objects, because the US military did not publish any positional information for the XSS-11 while on orbit or since.

In 2007, DARPA conducted a demonstration of RPO technology in the context of satellite servicing with its Orbital Express mission. Orbital Express consisted of two spacecraft, the ASTRO servicing vehicle and the NEXTSat client vehicle that were launched together into a roughly 500 km circular orbit. ASTRO demonstrated the ability to autonomously transfer fluid to NEXTSat and use a robotic arm to swap out components. The two spacecraft then separated and spent the next few months demonstrating multiple rendezvous and capture scenarios, including the first-ever use of a robotic arm to autonomously capture another space object. While the NEXTSat and ASTRO mission were primarily conducted for servicing research, its directive by DARPA leads to a possible dual-use, military RPO application. The two spacecraft were deactivated in July 2007.[3]

US RPO FOR MILITARY AND INTELLIGENCE MISSIONS IN GEO

The earliest known example of US RPO for military and intelligence missions is a satellite reportedly called Prowler.[4] According to analysis done by hobbyist satellite observers, Prowler was stealthily deployed by a 1990 military Space Shuttle mission into GEO. Prowler likely used optical camouflage techniques to maneuver close to multiple Russian satellites to collect intelligence on their characteristics and capabilities while remaining undetected by Russian optical space surveillance systems. It is thought to have been decommissioned around 1998. To this day, the United States has never officially acknowledged the existence of Prowler and lists it as an extra rocket body from the Shuttle launch in its public satellite catalog.

In 2006, the USAF launched two small satellites into GSO, officially designated as Micro-satellite Technology Experiment (MiTEx), with the official mission to identify, integrate, test, and evaluate small satellite technologies to support and enhance future US space missions.[5] Observers speculated that MiTEx satellites would be conducting RPO in GSO.[6] In 2009, news reports revealed that MiTEx satellites were used to conduct "flybys" of the US early-warning satellite DSP 23, which mysteriously failed on-orbit shortly after launch.[7] Hobbyists' observations noted that two MiTEx satellites maneuvered from the parking slots in GSO to drift towards the location of DSP 23, passing it around December 23, 2009 and January 1, 2010.

In recent years, the USAF appears to have applied the lessons it learned with Prowler and MiTEx to an operational program known as the Geosynchronous Space Situational Awareness Program (GSSAP), which may have the internal codename of Hornet. GSSAP uses multiple pairs of small satellites deployed slightly above and below the GEO belt; these satellites drift east and west and provide close inspections of objects in the GEO region.[8] The first pair of GSSAP satellites was launched on July 28, 2014, and the second pair on August 19, 2016. A third pair was launched in January 2022.[9] The US Space Systems Command confirmed in August 2023 that one of the satellites from the pair launched in 2014, GSSAP Space Vehicle (SV) 2, had reached its end of life and been placed in a graveyard orbit.[10] The USSF has ordered at least two more GSSAPs (GSSAPs 7 and 8) from their manufacturer, Northrop Grumman.[11] Two more launches of new GSSAP satellites were planned to occur in 2024 and 2027; the 2024 launch did not happen.[12] GSSAPs 7 and 8 were launched via a United Launch Alliance Vulcan Centaur in February 2026.[13] The GSSAP satellites are operated by the 1st Space Operations Squadron of the USSF's Space Delta 9, which has a mission to conduct orbital warfare.[14]

On September 18, 2015, General John E. Hyten, then Commander of Air Force Space Command, remarked at a public forum that the two GSSAP satellites had been "pressed into early service" to provide information to an unnamed customer.[15] According to General Hyten, the two satellites provided what he deemed "eye-watering" pictures of one or more objects in GSO.

Although the US military did not initially provide any public data on the locations or maneuvers of the GSSAP satellites, other sources of tracking data show they are very active in the GEO region. Data collected by the ISON space surveillance network, managed by the Russian Academy of Sciences, indicates that the GSSAP satellites have conducted hundreds of maneuvers since 2014 and have conducted close approaches or proximity operations of more than a dozen operational satellites in GEO, including Russian and Chinese military satellites, as well as commercial satellites built by China and operated by other countries. US commercial space situational companies released a video demonstrating an August 2020 approach of USA 271 to China's SJ-20 and tracking data demonstrating that in January 2022, USAF 270 approached the SY-12 01 and SY-12 02.[16] A Chinese paper released in 2023 indicated that they had tracked GSSAP satellites conducting at least 14 uncoordinated close approaches with six Chinese satellites from 2020-2021.[17]

GSSAP satellites.
Image credit: US Air Force.

USA 271 (GSSAP 4) maneuvered along the GEO belt in April 2025 to conduct a close approach to the recently launched TJS-15, estimated to have been 33 km, according to COMSPOC.[18] Later that same month, COMSPOC announced that USA 324 (GSSAP 5) had done its own close approaches to two other recently launched Chinese satellites, getting within 17 km of TJS-16 and 12 km of TJS-17.[19]

The launch of the first two GSSAP satellites included a satellite from another RPO program, the Automated Navigation and Guidance Experiment for Local Space (ANGELS) Program.[20] The goal of ANGELS was to provide a clearer picture of the local area around important US national security satellites in GEO. The first ANGELS satellite stayed attached to the Delta 4 upper stage after it conducted a disposal maneuver to place it a few hundred kilometers above GEO. At that point, ANGELS detached from the upper stage and conducted a series of RPO maneuvers to close within a few kilometers.[21] Russian tracking sources indicate that during one close approach conducted on June 9, 2016, the Delta upper stage altered its orbit, suggesting it might not have been totally inert. ANGELS was decommissioned in November 2017.[22]

On April 14, 2018, the United States conducted another military launch that placed multiple small satellites in GEO, including at least one that has conducted rendezvous and proximity operations.[23] Alongside the USAF's Continuous Broadcast Augmenting SATCOM (CBAS) military communication relay satellite, the launch also included the Evolved Expendable Launch Vehicle (EELV) Secondary Payload Adapter (ESPA) Augmented Geosynchronous Laboratory Experiment satellite, known by the triple-nested acronym EAGLE but officially cataloged as USA 284. The ESPA ring is commonly used for deploying small satellites as secondary payloads, and the EAGLE concept converts the ESPA ring from part of the launch vehicle into an independent maneuverable satellite, allowing for more flexible deployment of multiple small satellites.[24]

On this first launch, the EAGLE separated from the upper stage in the GEO region and subsequently deployed at least three small satellites. One of these small satellites, Mycroft, separated from EAGLE in early May 2018 and conducted a series of close approaches to EAGLE. The USAF describes it as demonstrating "improved space situational awareness for space vehicles."[25] In January 2020, the US military began providing public orbital information for CBAS, the Centaur upper stage, and the other two unnamed payloads, but not EAGLE or Mycroft.[26]

In October 2019, the USAF announced that Mycroft was being sent to inspect another US satellite in the GEO region, S5.[27] S5 was an experimental satellite launched into GEO on February 22, 2019, to test new space situational awareness concepts, but it stopped communicating with ground controllers in March 2019.[28] The USAF stated that Mycroft would conduct a series of RPO maneuvers with S5 over a period of weeks to try and determine the status of the latter's solar panels.[29] Amateur observers noted that Mycroft was communicating using a largely "suppressed" carrier signal, making it more difficult to detect.[30]

Three additional mysterious satellites are also conducting RPO activities in GEO. Known as Palladium at Night (PAN), Mentor-4, and CLIO, they were launched in 2009 (PAN and Mentor-4) and September 2014 (CLIO; not much is known about CLIO, other than that it is thought to be a second PAN, has maneuvered between orbital slots "multiple times").[31] The Mentor-4, or USA 202,[32] was launched on January 18, 2009, to a near-geosynchronous orbit at the longitude of 100 degrees east. Mentor-4 drifted westward around 0.8 degrees per day for several weeks, and eventually stabilized in a geosynchronous orbit at 44 degrees east longitude in May 2009.[33] That placement put it right next to commercial satellite Thuraya 2, where it appeared to match up with the latter's orbital inclination. In 2016, it appeared that Mentor-4 was conducting closely synchronized co-orbital movements with Thuraya 2.[34]

A classified satellite, PAN, was launched on September 8, 2009 into GEO, where it was observed relocating every six months or so, until late 2013. Very little is known about the mission of PAN, although most public observers believe it has a signals intelligence mission and that it is thought to be part of a series called Nemesis. Leaked files in 2016 stated its mission was "Foreign Satellite (FORNSAT) collection from space – targeting commercial satellite uplinks not normally accessible via conventional means"[35] and could be conducting similar activities to the Russian Luch/Olymp-K satellite. The PAN satellite moved roughly every six months during its first four years of life; those nine moves over four years placed it near several other satellites.[36] PAN stayed in a stable position until roughly February 2021, when it moved again.[37] Amateur observations noted that PAN moved in August 2023.[38] In September 2023, PAN was spotted drifting westward[39] and in April 2024, it drifted westward at an average of about 0.03 degrees longitude daily.[40]

The available information suggests these satellites are likely being used for signals intelligence (SIGINT) collection of broadcasts being sent to other communications satellites. Their behavior and likely mission is very similar to the Russian Luch/Olymp-K satellite that conducted similar activities in GEO from 2014 until it was put in a graveyard orbit in 2025.

France and the United States held their first bilateral RPO engagement in April 2025, which involved a US and a French satellite approaching each other near a "strategic competitor spacecraft;" it is unclear which country that spacecraft belonged to or which US and French satellites conducted the RPO.[41] In September 2025, a USSF official said that France and the United States would be engaging in a second RPO mission.[42] This ended up happening as part of Operation Olympic Defender in November 2025.[43] While both countries acknowledged the maneuvers, they did not provide specifics. However, COMSPOC monitored the RPOs and identified the participating satellites as USA 324 (part of the United States' GSSAP program) and France's SYRACUSE 3A; there were three sets of maneuvers observed (November 11-14, November 22-23, and November 28-29), and, according to a COMSPOC spokesperson, "In all these movements, SYRACUSE 3A seems to lead and USA 324 seems to follow as the maneuvers performed by USA 324 is lagged by a day."[44] The closest the two satellites got to each other was just a little over 25 km.[45]

In September 2025, US Space Command and UK Space Command announced the conduction of their first jointly coordinated RPO maneuver, occurring between September 4-12, 2025.[46] During this RPO maneuver, UK Space Command stated that a satellite “maneuvered near the UK SKYNET 5A communications satellite, a complex and meticulously planned operation.”[47] Going into more detail, COMSPOC reported that it was one of the US GSSAP satellites, USA 271, which had been drifting westward but stopped near SKYNET-5A at 95.3°E from September 5-11, 2025.[48] USA 271 had been drifting since July 2025 at a rate of -1.5° per day, until an in-track maneuver was detected on September 4, 2025, with additional maneuvers that followed, until USA 271 was within 0.05° of the British satellite and, at its closest point, was 13 km away.[49] According to Major General Paul Tedman, commander of UK Space Command, “This operation was a first of its kind for U.K. Space Command.”[50]

An unnamed space-based SSA program with three novel sensor payloads developed by the Space Rapid Capabilities Office (SpRCO) was launched in January 2023 by the USSF; according to the SpCRO, the payloads “fly around GEO and collect data about potential threats on-orbit.”[51] The payloads are likely attached to the Long Duration EELV Secondary Payload Adapter, LDPE 3A, which hosts two other technology demonstration payloads for the USSF.[52] In October 2024, LDPE 3A maneuvered close to China’s SJ-23, possibly getting within 30 km of it on November 4 and 5. After this, it maneuvered to synchronize itself with SJ-23 and drift along with it 0.68 degrees west per day.[53] Some payloads on the LDPE 3A have been collecting information about China’s space situational awareness capabilities and, according to Kelly Hammett of the Space Rapid Capabilities Office, “are sensors that can tell whether you’re being observed, tracked, targeted.”[54]

Another bus in that same series, LDPE 2, was detected releasing satellites in January 2023 (USA 341), April 2023 (USA 344), August 2024 (USA 399), and June 2025 (USA 546).[55] According to its manufacturer, the platform’s “six payload ports are capable of accommodating any combination of up to six hosted and 12 separable (fly-away) payloads (maximum one hosted or two separables per port).”[56]

The Space Force is striving to provide capabilities under its tactically responsive space (TacRS) program. Some of the program to date has focused on launching satellites very quickly on demand, which is not in and of itself a counterspace capability. The upcoming versions of this program are intended to provide RPO capabilities, which could provide on-orbit counterspace capabilities. Victus Haze was intended to launch in fall 2025; during the exercise, Rocket Lab was intended to launch a pre-manufactured payload 24 hours after it received the notification to do so and conduct RPO activities with True Anomaly’s Jackal Autonomous Orbital Vehicle spacecraft (launched in December 2024).[57] Victus Haze’s launch got delayed to some point in 2026 due to Firefly’s 2025 rocket anomalies.[58] In October 2024, Impulse Space won a $34.5 million contract as part of the TacRS program to create two two orbital maneuver vehicles, Victus Surgo and Victus Salo; Victus Surgo is supposed to move a payload from LEO to GEO, while Victus Salo would carry what has been only described as a “government-provided payload.”[59] Also part of this series is Victus Sol, awarded to Firefly in February 2025, which is intended to put an operational capability in orbit, but the specifics have been kept classified.[60]

The most likely military utility of the capabilities demonstrated by satellites like the DART, XSS-10, XSS-11, Orbital Express, Prowler, MiTEx, PAN, Mentor 4, GSSAP, ANGELS, Mycroft, and LDPE 2 and 3A are on-orbit SSA and close-up inspections. Their operational pattern is consistent with relatively slow and methodical approaches to rendezvous with other space objects in similar orbits. The satellites they are known to approach were in similar orbits and, based on publicly available data, they did not make large changes to rendezvous with satellites in significantly different orbits. This behavior is similar to several international RPO missions to test and demonstrate satellite inspection and servicing capabilities, in particular the Chinese SJ-12, SJ-15, SJ-17, and TJS-3 satellites.

Summary Of Known Or Suspected US Military And Intelligence RPO Activities In Space

Date of Test	Target Satellite	Chaser Satellite	Notes
1990 - 1998	Unknown communications satellites	Prowler	Maneuvered close to multiple Russian geosynchronous orbit satellites to collect intelligence but utilized optical stealth technologies to remain undetected.
Jan. 2003	Delta upper stage	XSS-10	XSS-10 did a series of maneuvers to bring it within 50 meters of the Delta upper stage that placed it in orbit.
Apr. 2005 - Oct. 2006	Minotaur upper stage, multiple US space objects	XSS-11	XSS-11 did a series of maneuvers to bring it close to the Minotaur upper stage that placed it in orbit. It then performed additional close approaches to other US space objects in nearby LEO orbits over the next 12-18 months.
Apr. 2005	MUBLCOM	DART	DART conducted a series of autonomous maneuvers to bring it close to the MUBLCOM satellite and ended up accidentally bumping into it.
Mar. - Jul. 2007	NEXTSat	ASTRO	ASTRO and NEXTSat were launched together and performed a series of separations, close approaches, and dockings with each other, including the first-ever use of a robotic arm to autonomously capture another space object. ASTRO also demonstrated the ability to autonomously transfer fluid to NEXTSat.
Dec. 23, 2008 and Jan. 1, 2009	DSP-23	MiTEx (USA 187, USA 188)	Inspection and close rendezvous with a failed US satellite. Possibly other demonstrations and tests in geosynchronous orbit.
2009 - Present	Yahsat 1B, others unknown	PAN (USA-207)	Part of NRO's Nemesis satellites. Presumed to be conducting SIGINT (signals intelligence) by parking next to other satellites. It was most active between 2009 and 2013, when it would move roughly every six months. It is still on orbit today but is mostly drifting in GEO.
2009 - Present?	Thuraya-2	Mentor 4 (USA-202)	Presumed to have conducted SIGINT (signals intelligence) by parking next to Thuraya-2.
Sept. 2014 - Present	Unknown	CLIO (USA 257)	Clio has approached other orbital slots "multiple" times.
Jul. 2014 - Present	Multiple satellites	GSSAP	Multiple pairs of GSSAP satellites have been performing RPO with other satellites in GEO; see the next table for specifics.
Jul. 2014 - Nov. 2017	Delta 4 R/B	ANGELS (USA 255)	ANGELS separated from the Delta 4 upper stage that placed the first GSSAP pair into orbit and then performed a series of RPO in the GSO disposal region.
May 2018	EAGLE	MYCROFT	EAGLE separated from the Delta V upper stage, and Mycroft subsequently separated from EAGLE. Mycroft conducted RPO of EAGLE in the GEO region.
Oct. 2019	S5	MYCROFT	Mycroft maneuvered to rendezvous with S5 after the latter ceased communications.
Aug. 2020	SJ-20	USA 271	USA 271 approached China's SJ-20, shadowing the spacecraft. The Chinese spacecraft detected the US satellite and rapidly moved away.
Jan. 2022	Shiyan-12 (SY-12 01) and Shiyan-12 (SY-12 02)	USA 270	USA 270 approached China's Shiyan-12 (SY-12 01) and (SY-12 02) satellites in GEO. As USA 270 approached, the Shiyan-12 satellites maneuvered away to drift orbits. The closest approach was around 73 kilometers.

Date of Test	Target Satellite	Chaser Satellite	Notes
Oct. 2024	SJ-23	LDPE 3A	In October 2024, LDPE 3A maneuvered to approach SJ-23, possibly to 30 km by Nov. 4 or Nov. 5, after which it maneuvered to synchronize itself with SJ-23.
Apr. 2025	TJS-15	USA 271	USA 271 maneuvered to conduct a close approach to the recently-launched TJS-15, and is estimated to have gotten within 33 km of it.
Apr. 2025	TJS-16 and TJS-17	USA 324	USA 324 maneuvered to conduct close approaches to two other recently launched Chinese satellites, getting within 17 km of TJS-16 and 12 km of TJS-17.
Apr. 2025	US satellite, French satellite	US satellite, French satellite	A US and French satellite approached each other near a “strategic competitor spacecraft;” it is unclear which country that spacecraft belonged to or which US and French satellites conducted the RPO.
Sept. 2025	SKYNET-5A	USA 271	USA 271 had been drifting since July 2025 at a rate of -1.5° per day, until an in-track maneuver was detected on September 4, 2025. Additional maneuvers followed, until USA 271 was within 0.05° of SKYNET-5A and, at its closest point, was 13 km away. USA 271 and SKYNET 5A’s RPO lasted roughly from Sept. 5-11, 2025.
Nov. 2025	SYRACUSE 3A	USA 324	There were three sets of maneuvers observed between the US and French satellites (November 11-14, November 22-23, and November 28-29), and, according to a COMSPOC spokesperson, “In all these movements, SYRACUSE 3A seems to lead and USA 324 seems to follow as the maneuvers performed by USA 324 is lagged by a day.” The closest the two satellites got to each other just a little over 25 km.

Satellites Approached By GSSAP Through January 2022[61]

Date	Satellite Approached	Country of Ownership	Approach Distance
Sept. 13, 2016	TJS-1	China	15 km
Jul. 13, 2017	Express AM-8	Russia	10 km
Sept. 14, 2017	Luch	Russia	10 km
Sept. 21, 2017	Paksat 1R	Pakistan	12 km
Sept. 29, 2017	Nigcomsat 1R	Nigeria	11 km
Oct. 5, 2017	Blagovest (Cosmos 2520)	Russia	14 km
Nov. 17, 2017	Raduga-1M 3	Russia	12 km
May 14, 2018	Raduga-1M 2	Russia	13 km
Feb. 26, 2020	Tianlian 2-01	China	88 km
Feb. 28, 2020	BD-2 G8	China	110 km
Apr. 8, 2020	SJ-13	China	360 km
Aug. 23, 2020	SJ-20/Chinasat 6A	China	24 km
Nov. 1, 2020	SJ-20	China	12 km
Nov. 23, 2020	SJ-13	China	102 km
Nov. 30, 2020	TJS-2	China	53 km
Jan. 1, 2021	TJS-3 AKM	China	469 km
Mar. 19, 2021	TJS-2	China	28 km
Apr. 16, 2021	TJS-3	China	1141 km
May 3, 2021	SJ-20	China	1032 km
May 4, 2021	SJ-20	China	10 km
May 6, 2021	TJS-3	China	104 km
Jan. 2022	SY-12 01, SY-12 02	China	73 km

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